MAT 2040 Linear Algebra
Assignment 2

Released date: 2024/09/23.
Due: 2024/10/08.
Late submission is NOT acceptable.
Please submit your assignment as a PDF file titled "student ID + HW2".

Question 1

The matrix

$$A = \begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix}$$

has the property that $A^2 = O$. Is it possible for a nonzero symmetric 2×2 matrix to have this property? Prove your answer.

Question 2

Let C be a nonsymmetric $n \times n$ matrix. For each of the following, determine whether the given matrix must necessarily be symmetric or could possibly be nonsymmetric:

(a) $A = C + C^T$

(b) $B = C - C^T$

(c) $D = C^T C$

(d) $E = C^T C - C C^T$

(e) $F = (I + C)(I + C^T)$

(f) $G = (I + C)(I - C^T)$

Question 3

Let

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$$

Show that if $d = a_{11}a_{22} - a_{21}a_{12} \neq 0$, then

$$A^{-1} = \frac{1}{d}\begin{pmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{pmatrix}$$

Question 4

Let A and B be $n \times n$ matrices. Show that if $AB = A$ and $B \neq I$, then A must be singular.

Question 5

Prove that if A is nonsingular then A^T is nonsingular and $(A^T)^{-1} = (A^{-1})^T$. Hint:$(AB)^T = B^T A^T$

Question 6

Let A be an $m \times n$ matrix. Show that $A^T A$ and AA^T are both symmetric.

Question 7

Let

$$A = \begin{pmatrix} 1 & 2 & 4 \\ 2 & 1 & 3 \\ 1 & 0 & 2 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 2 & 4 \\ 2 & 1 & 3 \\ 2 & 2 & 6 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 2 & 4 \\ 0 & -1 & -3 \\ 2 & 2 & 6 \end{pmatrix}$$

(a) Find an elementary matrix E such that $EA = B$.

(b) Find an elementary matrix F such that $FB = C$.

(c) Is C row equivalent to A? Explain.

Question 8

Let

$$A = \begin{pmatrix} 2 & 1 & 1 \\ 6 & 4 & 5 \\ 4 & 1 & 3 \end{pmatrix}$$

(a) Find elementary matrices E_1, E_2, E_3 such that $E_3E_2E_1A = U$, where U is an upper triangular matrix.

(b) Determine the inverses of E_1, E_2, E_3 and set $L = E_1^{-1}E_2^{-1}E_3^{-1}$. What type of matrix is L? Verify that $A = LU$.

Question 9

Let

$$A = \begin{pmatrix} 2 & 1 \\ 6 & 4 \end{pmatrix}$$

(a) Express A^{-1} as a product of elementary matrices.

(b) Express A as a product of elementary matrices.

Question 10

Compute the LU factorization of each of the following matrices:

(a) $\begin{bmatrix} 3 & 1 \\ 9 & 5 \end{bmatrix}$

(b) $\begin{bmatrix} 2 & 4 \\ -2 & 1 \end{bmatrix}$

(c) $\begin{bmatrix} 1 & 1 & 1 \\ 3 & 5 & 6 \\ -2 & 2 & 7 \end{bmatrix}$

(d) $\begin{bmatrix} -2 & 1 & 2 \\ 4 & 1 & -2 \\ -6 & -3 & 4 \end{bmatrix}$

Question 11

Find the inverse of each of the following matrices:

(a) $\begin{pmatrix} -1 & 1 \\ 1 & 0 \end{pmatrix}$

(b) $\begin{pmatrix} 2 & 5 \\ 1 & 3 \end{pmatrix}$

(c) $\begin{pmatrix} 2 & 6 \\ 3 & 8 \end{pmatrix}$

(d) $\begin{pmatrix} 3 & 0 \\ 9 & 3 \end{pmatrix}$

(e) $\begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$

(f) $\begin{pmatrix} 2 & 0 & 5 \\ 0 & 3 & 0 \\ 1 & 0 & 3 \end{pmatrix}$

(g) $\begin{pmatrix} -1 & -3 & -3 \\ 2 & 6 & 1 \\ 3 & 8 & 3 \end{pmatrix}$

(h) $\begin{pmatrix} 1 & 0 & 1 \\ -1 & 1 & 1 \\ -1 & -2 & -3 \end{pmatrix}$

Question 12

Let A be a 3×3 matrix and suppose that

$$2a_1 + a_2 - 4a_3 = 0$$

How many solutions will the system $Ax = 0$ have? Explain. Is A nonsingular? Explain.

Question 13

Show that if A is a symmetric nonsingular matrix, then A^{-1} is also symmetric.

Question 14

Prove that B is row equivalent to A if and only if there exists a nonsingular matrix M such that $B = MA$.

Question 15

Perform each of the following block multiplications:

(a)

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & -1 \\ 2 & 1 & 2 & -1 \end{array}\right] \left[\begin{array}{ccc} 4 & -2 & 1 \\ 2 & 3 & 1 \\ 1 & 1 & 2 \\ \hline 1 & 2 & 3 \end{array}\right]$$

(b)

$$\left[\begin{array}{cc} 4 & -2 \\ 2 & 3 \\ 1 & 1 \\ \hline 1 & 2 \end{array}\right] \left[\begin{array}{ccc|c} 1 & 1 & 1 & -1 \\ 2 & 1 & 2 & -1 \end{array}\right]$$

(c)

$$\left[\begin{array}{cc|cc} \frac{3}{5} & -\frac{4}{5} & 0 & 0 \\ \frac{4}{5} & \frac{3}{5} & 0 & 0 \\ \hline 0 & 0 & 1 & 0 \end{array}\right] \left[\begin{array}{cc|c} \frac{3}{5} & \frac{4}{5} & 0 \\ -\frac{4}{5} & \frac{3}{5} & 0 \\ \hline 0 & 0 & 1 \\ 0 & 0 & 0 \end{array}\right]$$

(d)

$$\left[\begin{array}{ccc|cc} 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{array}\right] \left[\begin{array}{cc} 1 & -1 \\ 2 & -2 \\ 3 & -3 \\ \hline 4 & -4 \\ 5 & -5 \end{array}\right]$$

Question 16

Let A be an $m \times n$ matrix, X an $n \times r$ matrix, and B an $m \times r$ matrix. Show that

$$AX = B$$

if and only if

$$Ax_j = b_j, \quad j = 1, \ldots, r$$

Question 17

Let

$$A = \begin{pmatrix} A_{11} & A_{12} \\ O & A_{22} \end{pmatrix}$$

where all four blocks are $n \times n$ matrices.

(a) If A_{11} and A_{22} are nonsingular, show that A must also be nonsingular and that

$$A^{-1} = \left[\begin{array}{c|c} A_{11}^{-1} & C \\ \hline O & A_{22}^{-1} \end{array}\right]$$

(b) Determine C.